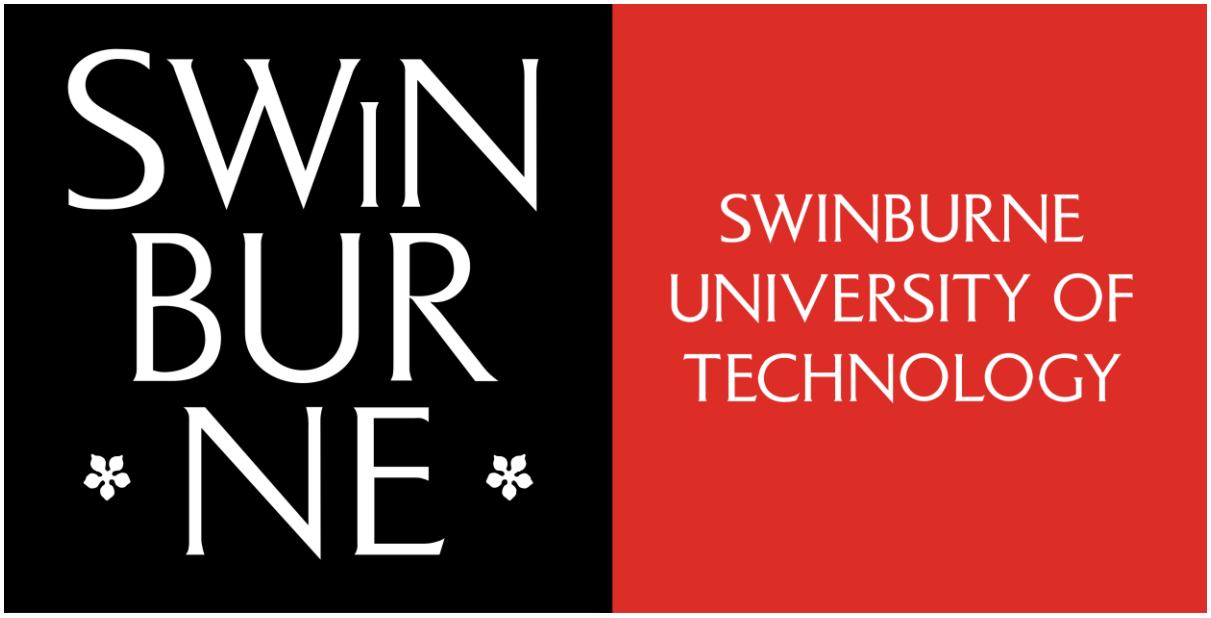
COS80023 Research Report

Are female occupants more likely to be seriously injured than male occupants in comparable crashes in Victoria, and has newer vehicle design closed or widened that gap?

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Summary

This project investigated whether female vehicle occupants are more likely than male occupants to be killed or seriously injured in comparable crashes in Victoria, and whether newer vehicle design has narrowed or widened this gap. Vehicle safety features have historically been developed and tested using crash test dummies based on the average male body, so understanding whether women remain at higher risk can help inform vehicle design, safety testing standards and road safety policy.

We used the Victorian Road Crash Data from January 2012 to September 2024, which combines police crash records with hospital injury information. Using R, we joined the person, vehicle and accident tables and identified 194,061 seatbelted drivers and front-seat passengers of cars, station wagons and utes for exploratory analysis. We compared KSI rates across recorded characteristics and grouped similar crashes using k-modes clustering. The final adjusted logistic regression used 175,453 occupants aged 16 or older with complete modelling data. A separate predictive analysis compared logistic regression with random forest.

After adjustment for recorded occupant, vehicle and crash characteristics, estimated KSI probabilities were 18.87% for women and 14.65% for men, a difference of 4.22 percentage points and a marginal odds ratio of 1.36. Adjustment for vehicle weight did not remove the difference. The gap was larger among front-seat passengers than drivers and persisted in several crash configurations, including rear-end, head-on and intersection-related crashes; run-off-road and loss-of-control comparisons were inconclusive. Newer vehicles were associated with lower KSI probabilities for both sexes, but the analysis did not establish whether the gap narrowed or widened with manufacture year. Random forest offered only small predictive improvements over logistic regression. Overall, the findings indicate a persistent adjusted association, not proof of its causes.

The data lacked occupant body measurements, restraint-fit information and detailed collision-severity measures, limiting explanations of the gap. Only crashes in which someone was injured were included. Seatbelt status was unknown for 28.3% of otherwise eligible occupants with recorded sex, and including this group reduced the descriptive gap; an adjusted sensitivity analysis of these exclusions was not completed. Manufacture year was only a proxy for safety design. Future research could link occupant measurements and vehicle safety ratings, assess missing-data effects and validate predictions on external or later-period data.

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1. Introduction

Modern cars are far safer than they were a few decades ago, but that progress may not have been shared equally. For many years, vehicle safety features such as seatbelts, airbags and seat designs were developed and tested using crash test dummies modelled on the average adult male. Women are, on average, shorter and lighter, sit closer to the steering wheel, and have different body proportions, which raises the question of whether cars protect them as well as they protect men. A US study found that a seatbelted female driver had 47% higher odds of severe injury than a seatbelted male driver in a comparable crash (Bose, Segui-Gomez & Crandall 2011). Less is known about whether the same pattern exists on Australian roads, or whether it has changed as vehicles have become safer.

This project explores that question using the Victorian Road Crash Data, a large dataset of police-reported crashes linked with hospital injury information. We focused on seatbelted drivers and front-seat passengers of cars, station wagons and utes, so that men and women were compared in similar situations. We first compared serious injury rates across vehicle age, weight, seating position and crash type, then used clustering to group similar crashes, and finally built statistical and machine learning models to account for several factors at once.

Our aim was to find out whether women are more likely than men to be killed or seriously injured in comparable crashes, and whether newer vehicle design has narrowed or widened that gap.

2. Methodology

Overview

This project used three of the nine tables in the Victorian Road Crash Data: the person, vehicle and accident tables, covering crashes from January 2012 to September 2024. All work was carried out in R. Initially, we joined the three tables into a single dataset with one row per person, linking each person to the vehicle they were in and the crash they were involved in. We also added information about the other vehicles in each crash, such as the weight of the heaviest other vehicle. We then filtered the data in stages to seatbelted drivers and front-seat passengers of cars, station wagons and utes with a recorded sex, recording the number of people remaining after each step. This produced a final dataset of 194,061 occupants.

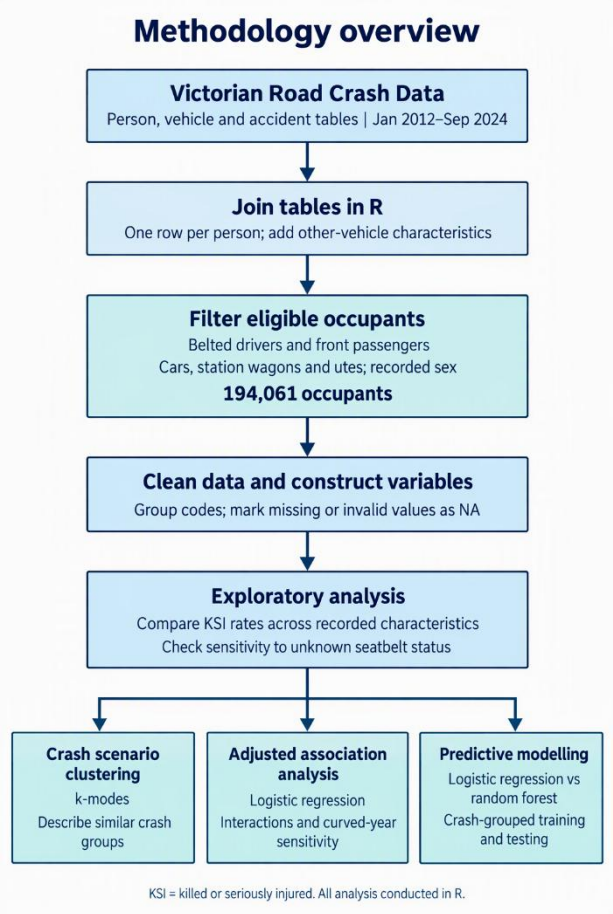


Figure 1. Methodology Overview

Next, we cleaned the data and grouped raw codes into clear categories, such as vehicle era, vehicle weight quartile, age band, speed zone and point of impact. Missing or invalid values were marked as missing rather than removed. We then carried out exploratory analysis, comparing the proportion of women and men killed or seriously injured across one factor at a time, and tested whether excluding occupants with unknown seatbelt status affected the results.

Finally, we grouped similar crashes into scenarios using k-modes clustering, and fitted logistic regression models to estimate the difference between women and men while accounting for several factors at once. A random forest model was also trained and compared with logistic regression to assess how well serious injury could be predicted.

Data Preparation

Data sources and tools

The analysis used the person, vehicle and accident tables from the Victorian Road Crash Data. The person table contains one row per person involved in a crash (425,352 rows), the vehicle table one row per vehicle (331,992 rows), and the accident table one row per crash (182,039 rows). Codes and their meanings were checked against the official data dictionaries published by the Victorian Department of Transport and Planning. The project was originally planned in Hive on Azure HDInsight, but because access to the unit's Azure subscription was unavailable, and with our tutor's approval, all data preparation was carried out in R using the dplyr and readr packages.

All columns were first read as text, so that codes with leading zeros (for example, vehicle type “01” or speed zone “060”) were not changed into numbers. Numeric columns such as vehicle year and tare weight were then converted. A tare weight of 0 kg was treated as missing, since it is not a valid vehicle weight. Checks confirmed there were no duplicate records in any of the three tables.

Joining the tables

Each person was matched to their vehicle using the accident number and vehicle ID, and to their crash using the accident number. People without a vehicle, mostly pedestrians, were removed at this step. Before joining, we calculated three features describing the other vehicles in the same crash: the number of other vehicles, the tare weight of the heaviest other vehicle, and whether any other vehicle was a truck, bus or light commercial vehicle. These were added so that later analysis could account for what each vehicle collided with.

Filtering

The joined data was filtered in five steps, with the number of people recorded after each step (Table 1).

Table 1: Filtering steps and number of occupants remaining

Step	Filter	Remaining	Removed	Reason
0	All people in the person table	425,352	–	–
1	Matched to a vehicle and crash	408,264	17,088	Pedestrians and others without a vehicle are not occupants
2	Drivers and left-front passengers	363,984	44,280	The study focuses on front-seat occupants
3	Cars, station wagons and utes	288,823	75,161	Light passenger vehicles with comparable safety design
4	Sex recorded as male or female	282,399	6,424	Sex is the main variable of interest
5	Seatbelt worn	194,061	88,338	Removes the large effect of seatbelt use on injury

Restricting the data to seatbelted occupants was an important decision. Seatbelt use strongly affects injury severity, so including unbelted occupants could make a difference in seatbelt use appear to be a difference between women and men. However, seatbelt status was recorded as unknown for 79,789 of the 282,399 occupants at Step 4. To check whether excluding them affected our findings, a second dataset including both belted occupants and those with unknown seatbelt status (273,850 occupants) was kept for a sensitivity analysis.

```
> source("~/Documents/GitHub/Big Data Final Project/R/01_prepare_data.R")
Rows read - person: 425352 | vehicle: 331992 | accident: 182039
Step 0  | All people in person table                                | 425,352 rows
Step 1  | Joined to a vehicle and crash                             | 408,264 rows
Step 2  | Drivers and front passengers                              | 363,984 rows
Step 3  | Cars, station wagons and utes                             | 288,823 rows
Step 4  | Sex recorded as M or F                                    | 282,399 rows

Seatbelt codes before the last filter:
# A tibble: 9 × 2
  helmet_belt_worn      n
  <chr>             <int>
1 1                194061
2 9                 79789
3 2                  5300
4 8                  1988
5 5                   982
6 3                   258
7 4                    19
8 6                     1
9 NA                    1
Step 5  | Seatbelt worn (FINAL TABLE)                               | 194,061 rows
Sensitivity table (belt worn or not known): 273,850 rows

KSI rate by sex (final table):
# A tibble: 2 × 4
  sex   people ksi_count ksi_pct
  <chr>  <int>     <int>   <dbl>
1 F      95942     17640    18.4
2 M      98119     14466    14.7

Saved:
  data/processed/occupants_final.csv
  data/processed/occupants_sensitivity.csv
  outputs/tables/step3_filter_counts.csv
>
```

Figure 2. R output showing the row count after each filtering step

Notes on the data

The dataset only includes crashes where at least one person was injured. However, uninjured occupants in those crashes are still recorded, which allows injury rates to be compared. For about 96% of vehicles, the number of person records matched the recorded number of occupants. Although the data dictionary describes the sex field as “the driver’s sex”, it is filled in for all people, including passengers.

Data Cleaning and Variable Construction

Approach

The final dataset from the preparation stage contained raw codes that were difficult to compare directly, such as individual manufacture years, exact vehicle weights and numbered impact codes. We grouped these into a small number of meaningful categories, which makes patterns easier to see in charts and gives the models a manageable number of groups to compare.

Missing, unknown or impossible values were set to missing (NA) rather than removed. This kept all 194,061 occupants in the dataset, so that each later analysis could use every occupant who had the values it needed. The outcome variable, killed or seriously injured (KSI), was coded as 1 if the injury level was fatal or serious and 0 otherwise. Sex and seating position were given readable labels, with male set as the reference group so that results compare women against men.

Variables created

Table 2 shows each grouped variable, how it was defined and how many values were set to missing.

Table 2: Grouped variables and missing values

Variable	Groups	Set to missing	Reason
Vehicle era	Pre-2000, 2000–2009, 2010–2015, 2016+	2,597	492 blank; 2,105 with impossible years (e.g. 0) or years before 1960 or after 2025
Vehicle weight (tare) quartile	Four equal-sized groups, from lightest (up to 1,243 kg) to heaviest (1,662 kg and above)	3,019	2,907 blank or 0 kg; 112 below 600 kg or above 3,500 kg, which is unrealistic for these vehicle types
Age band	16–25, 26–39, 40–59, 60+	4,093	3,544 children under 16, whose body size differs greatly from adults; 549 unknown age
Speed zone	50 km/h or less, 60-70, 75-90, 100-110	5,501	Codes for “other”, “off road” and “not known”
Crash type	Collision with vehicle, hit fixed object, rollover, other	0	The three most common crash types kept; the rest grouped as other
Point of impact	Front, side, rear, other	4,753	Codes for “not known”, towed unit or sidecar
Crash vehicles	Single vehicle, multi-vehicle	0	Based on the number of other vehicles in the crash

Vehicle era was used as an indicator of vehicle safety design, since newer vehicles generally include more advanced safety features. Weight quartiles were used to check whether any difference between women and men could be explained by women driving lighter vehicles.

Completeness

After cleaning, 175,500 occupants (90.4%) had a value for every grouped variable. This was considered high enough for modelling. The cleaned dataset was saved in R’s RDS format, which keeps the order of each category (for example, vehicle eras from oldest to newest), as well as in CSV format.

```
> # ---- Final check: how much is missing in each new column? -------------
> new_columns <- c("sex", "seat", "vehicle_era", "tare_quartile", "age_band",
+                  "speed_band", "crash_type", "impact", "crash_vehicles")
> colSums(is.na(occ[new_columns]))
           sex           seat    vehicle_era  tare_quartile       age_band
             0              0           2597           3019           4093
    speed_band     crash_type         impact crash_vehicles
          5501              0           4753              0
> # How many people have ALL of these filled in?
> # (This is roughly how many the models in Step 7 will be able to use.)
> sum(complete.cases(occ[new_columns]))
[1] 175500
> # ---- Save the clean table ---------------------------------------------
> # CSV = opens anywhere (Excel etc.)
> # RDS = R's own format; it remembers the category order we set with
> #       factor(), so it is the better one to load in later R scripts.
> write_csv(occ, "data/processed/occupants_clean.csv")

> saveRDS(occ,   "data/processed/occupants_clean.rds")
> cat("Saved occupants_clean.csv and occupants_clean.rds:", nrow(occ), "rows\n")
Saved occupants_clean.csv and occupants_clean.rds: 194061 rows
>
```

Figure 3. R output showing missing values in each grouped variable and the number of complete cases.

```
> # Check: how many people in each era, and how many missing (NA)?
> table(occ$vehicle_era, useNA = "ifany")

 Pre-2000 2000-2009 2010-2015     2016+      <NA>
    25233     82176     56354     27701      2597
>
```

Figure 4. Number of occupants in each vehicle era

Exploratory Analysis

Comparing injury rates

The exploratory analysis compared the percentage of women and men who were killed or seriously injured (the KSI rate) across one factor at a time. The factors examined were vehicle era, vehicle weight quartile, seating position, crash type, point of impact, and whether the crash involved a single vehicle or multiple vehicles. For each factor, occupants with a missing value for that factor were left out of that comparison only.

The difference between women and men was measured in two ways:

- **Gap in percentage points:** the female KSI rate minus the male KSI rate.
- **Ratio:** the female KSI rate divided by the male KSI rate. A ratio of 1.20 means women's rate was 20% higher than men's.

The ratio was the main measure, because it allows a fair comparison between groups with very different overall injury rates. For example, a 3-point gap matters more when the overall rate is 10% than when it is 40%.

To check whether women's higher rate could simply reflect the vehicles they drive, we also compared the median vehicle weight of women and men, and the share of each sex in each weight quartile.

Tools

The analysis used dplyr to calculate the rates and ggplot2 to produce grouped bar charts. To keep the analysis consistent and avoid repeating code, we wrote three reusable R functions: one to calculate KSI rates by sex for any factor, one to calculate the gap and ratio, and one to draw the chart. Each chart and summary table was saved automatically so that it could be included in the report. Blue and orange were chosen for men and women because they remain easy to tell apart for readers with colour blindness.

```
> # ---- 1. Overall: women vs men ----------------------------------------
> overall <- occ %>%
+   group_by(sex) %>%
+   summarise(people = n(), ksi_pct = round(100 * mean(ksi), 1))
> overall
# A tibble: 2 x 3
  sex    people ksi_pct
  <fct>   <int>   <dbl>
1 Male    98119    14.7
2 Female  95942    18.4
> # ---- 2. By vehicle era (the key question) ----------------------------
> era_rates <- rate_table(occ, "vehicle_era")
> era_gap   <- gap_table(era_rates)
> era_gap
# A tibble: 4 x 5
  group     women_pct men_pct gap_pts ratio
  <fct>         <dbl>   <dbl>   <dbl> <dbl>
1 Pre-2000       23.7    19.8     3.9  1.2
2 2000-2009      19.1    15.7     3.4  1.22
3 2010-2015      16.8    12.2     4.6  1.38
4 2016+          15.8    10.8     5    1.46
> era_chart <- plot_rates(era_rates,
+                         "Serious injury rate by sex and vehicle year of manufactur
e",
+                         "Vehicle year of manufacture")
```

Figure 5. R output showing the overall KSI rates and the gap table by vehicle era

Sensitivity analysis

Because 79,789 occupants with unknown seatbelt status were excluded during data preparation, we repeated the overall comparison using the second dataset, which included both belted occupants and those with unknown seatbelt status. We calculated the gap separately for each seatbelt group and for both groups combined. If the difference between women and men remained similar, this would show that our main finding was not caused by excluding these occupants.

```
> # ---- 9. Sensitivity check: include people with seatbelt unknown --------
> # This table (from Step 3) has seatbelt worn (code 1) AND not known (code 9)
> sens <- read_csv("data/processed/occupants_sensitivity.csv",
+                  col_types = cols(.default = "c")) %>%
+   mutate(
+     ksi = as.integer(ksi),
+     sex = factor(sex, levels = c("M", "F"), labels = c("Male", "Female")),
+     belt_group = ifelse(helmet_belt_worn == "1", "Seatbelt worn", "Seatbelt not kn
own"),
+     belt_group = factor(belt_group, levels = c("Seatbelt worn", "Seatbelt not know
n"))
+   )

> # Gap within each seatbelt group
> sens_gap <- gap_table(rate_table(sens, "belt_group"))
> sens_gap
# A tibble: 2 x 5
  group              women_pct men_pct gap_pts ratio
  <fct>                  <dbl>   <dbl>   <dbl> <dbl>
1 Seatbelt worn           18.4    14.7     3.7  1.25
2 Seatbelt not known      13.5    13.1     0.4  1.03
> # Gap with both groups combined (as if we had NOT dropped the unknowns)
> sens %>%
+   group_by(sex) %>%
+   summarise(people = n(), ksi_pct = round(100 * mean(ksi), 1))
# A tibble: 2 x 3
  sex    people ksi_pct
  <fct>   <int>   <dbl>
1 Male   143516    14.2
2 Female 130334    17.1
> write_csv(sens_gap, "outputs/tables/05_sensitivity_seatbelt.csv")
```

Figure 6. R output showing the sensitivity analysis.

Limitations of this approach

Each comparison looks at only one factor at a time, so it cannot separate the effects of factors that are related to each other. For example, if women's injury rate is higher in lighter cars, this could reflect either their sex or the car itself. The exploratory analysis was therefore used to identify patterns, which were then tested using regression models that account for several factors together.

Crash Scenario Clustering

We used k-modes clustering in R to group vehicles by five categorical characteristics: speed zone, broad crash type, initial impact location, intersection status and lighting (Neumann and Szepannek n.d.). Sex, KSI, occupant age, manufacture year and vehicle weight were excluded from the clustering inputs.

Each vehicle was counted once, regardless of how many eligible occupants it contained. Complete clustering data were available for 153,177 vehicles, representing 182,531 occupants. Vehicles with missing inputs remained unassigned rather than being removed from subsequent analyses.

We compared one to eight clusters on the same random sample of 10,000 vehicles, using repeated starts. We selected the number of clusters using the normalised elbow curve and checked separation using silhouette scores on a 2,000-vehicle subsample. We then refitted the selected solution to all complete vehicles, described the groups and compared observed KSI rates by sex. These comparisons were unadjusted. Figure 7 illustrates the clustering code and saved diagnostics; implementation settings are in Appendix D.

Crash scenario clustering

Source excerpt: R/04_cluster.R

```
features <- c("speed_band", "crash_type", "impact", "road_context", "lighting")
profile_features <- c(features, "crash_vehicles")
vehicle_rows <- occ %>%
  select(accident_no, vehicle_id, all_of(profile_features)) %>%
  distinct() %>%
  arrange(accident_no, vehicle_id)
if (anyDuplicated(vehicle_rows[c("accident_no", "vehicle_id")])) {
  stop("Occupants of the same vehicle have conflicting crash features.")
}

complete_vehicle <- complete.cases(vehicle_rows[features])
complete_occupant <- complete.cases(occ[features])
```

Saved analysis output

outputs/tables/06_cluster_filter_counts.csv

Stage	Vehicles	Occupants
Clean input	163141	194061
Complete clustering features	153177	182531
Unassigned: missing features	9964	11530

Code excerpts and saved outputs are reproduced directly from the project files.

Figure 7. Screenshot of clustering code and saved cohort counts

Regression Analysis

Logistic regression estimated associations with KSI, coded as 1 for fatal or serious injury and 0 otherwise. We used all 175,453 eligible occupants aged at least 16 with complete modelling data, representing 100,085 crashes. Required values had to be valid, crash configuration known, and manufacture year no later than crash year.

We grouped official DCA codes into seven crash configurations (Department of Transport and Planning n.d.). These were defined separately from clustering; cluster labels were not model inputs. The model included sex, manufacture year, tare weight, seating position, age band, speed band, crash configuration, impact location, vehicle count and crash year. Sex interactions with year, weight, seating, configuration and impact allowed the sex association to vary across these characteristics.

We used standard errors clustered by crash to account for related occupant outcomes. Joint Wald tests assessed the five interactions, with Holm adjustment for multiple testing (Zeileis et al. n.d.). For each comparison, we predicted KSI twice for the same reference occupants, setting sex first to male and then female. From the averaged probabilities, we calculated percentage-point gaps and marginal odds ratios, with pointwise 95% confidence intervals.

We checked the vehicle-year finding using a four-degree-of-freedom natural spline, allowing curved relationships by sex while retaining the other controls and interactions. These analyses estimate adjusted associations. Missing-data and exclusion sensitivity for the adjusted models was not completed. Figure 8 shows the final model code and saved checks; reference populations and technical settings are documented in Appendices A and D.

Final logistic regression

Source excerpt: R/16_impact_interaction.R

```
fit_model <- function(form, data) {
  fit <- glm(form, data = data, family = binomial(), na.action = na.fail,
             control = glm.control(maxit = 50))
  stopifnot(fit$converged, all(is.finite(coef(fit))), fit$rank == length(coef(fit)),
            nobs(fit) == nrow(cohort), identical(rownames(fit$model), rownames(cohort)))
  V <- sandwich::vcovCL(fit, cluster = keys$accident_no, type = "HC1", cadjust = TRUE)
  stopifnot(all(is.finite(V)), all(diag(V) > 0))
  list(model = fit, cluster_robust_vcov = V)
}
primary_formula <- update(formula(previous$model), . ~ . + sex:impact)
message("Fitting final linear-year association model with sex x impact")
primary <- fit_model(primary_formula, cohort)
```

Saved analysis output

outputs/tables/07_final_model_diagnostics.csv

Model	Occupants	Crashes	Parameters	Converged
Primary	175453	100085	46	TRUE
Curved	175453	100085	52	TRUE

Code excerpts and saved outputs are reproduced directly from the project files.

Figure 8. Screenshot of final regression code and saved model checks

Predictive Modelling

We separately compared logistic regression and random forest using the same cohort and recorded predictors. The logistic candidates used linear or curved manufacture-year relationships and preceded the final sex-by-impact interaction. This evaluation therefore does not validate the exact final association model.

We allocated 80% of crashes to training and 20% to testing, stratified by whether a crash included an eligible KSI occupant. Five-fold cross-validation also kept all occupants from each crash together. Within each model family, we selected the candidate with the lowest pooled out-of-fold log loss and refitted it using all training data. The forest used 300 trees (Wright n.d.); full tuning settings are in Appendix D.

We evaluated ranking, probability accuracy and calibration using ROC AUC, average precision, Brier score, log loss and calibration plots. Confusion matrices described classification errors. We compared a fixed 0.5 cutoff with training-derived thresholds based on Youden's criterion, at least 80% recall and maximum F1. These used out-of-fold training predictions and were applied unchanged to the test set; the recall target was illustrative.

This was retrospective internal validation. Earlier full-cohort exploration informed model choices, and the recall and F1 scenarios followed inspection of initial test results. No external or future-period validation was performed. Figure 9 illustrates the crash-grouped split and saved evaluation output.

Predictive validation by crash

Source excerpt: R/12_predictive_evaluation.R

```
43    assignments <- crashes %>% mutate(partition = "train", fold = NA_integer_)
44    for (stratum in 0:1) {
45      eligible <- which(assignments$any_ksi == stratum)
46      test_rows <- sample(eligible, size = round(0.2 * length(eligible)))
47      assignments$partition[test_rows] <- "test"
48      train_rows <- setdiff(eligible, test_rows)
49      assignments$fold[train_rows] <- sample(rep(1:5, length.out = length(train_rows)))
50    }
…
57  membership <- assignments[match(keys$accident_no, assignments$accident_no), ]
58  train_rows <- which(membership$partition == "train")
59  test_rows <- which(membership$partition == "test")
60  stopifnot(length(intersect(keys$accident_no[train_rows], keys$accident_no[test_rows])) == 0,
61            all(is.na(membership$fold[test_rows])), all(membership$fold[train_rows] %in% 1:5))
```

Saved analysis output

outputs/tables/07_predictive_cv_summary.csv

Candidate	Out-of-fold log loss	Selected
Logistic_linear	0.4011689334861719	TRUE
Logistic_curved	0.4012120102919281	FALSE
RF_mtry3	0.39884196129612864	TRUE
RF_mtry6	0.40183258297609503	FALSE

Code excerpts and saved outputs are reproduced directly from the project files.

Figure 9. Screenshot of crash-grouped validation code and saved output

3. Results

Overview

Female occupants had a higher probability of being killed or seriously injured (KSI) than male occupants in the analysed Victorian crashes. This difference remained after adjustment for recorded occupant, vehicle and crash characteristics. However, its size varied across crash circumstances and was greater among front-seat passengers than drivers. Newer and heavier vehicles were associated with lower estimated KSI probabilities for both sexes, but the analysis did not establish whether the sex gap narrowed or widened with vehicle manufacture year.

The exploratory analysis included 194,061 belted occupants. The adjusted analysis used 175,453 occupants aged 16 or older with complete modelling data, representing 100,085 crashes. The following sections address the four research sub-questions, with key tables and figures alongside the discussion. Appendices A to C contain the complete adjusted comparisons, prediction results and exploratory findings.

Exploratory findings

Before adjusting for other factors, we compared the observed KSI rates of women and men across one factor at a time. Overall, 18.4% of belted female occupants were killed or seriously injured, compared with 14.7% of male occupants, meaning women's rate was 1.25 times men's.

KSI rates fell steadily for both sexes in newer vehicles (Figure 10). However, men's rate fell further, from 19.8% in vehicles built before 2000 to 10.8% in vehicles built from 2016 onwards, while women's rate fell from 23.7% to 15.8%. As a result, the observed ratio between women and men increased from 1.20 to 1.46. This suggested that improvements in vehicle safety may have benefited men more than women, which was tested further in the adjusted analysis (Research question 2).

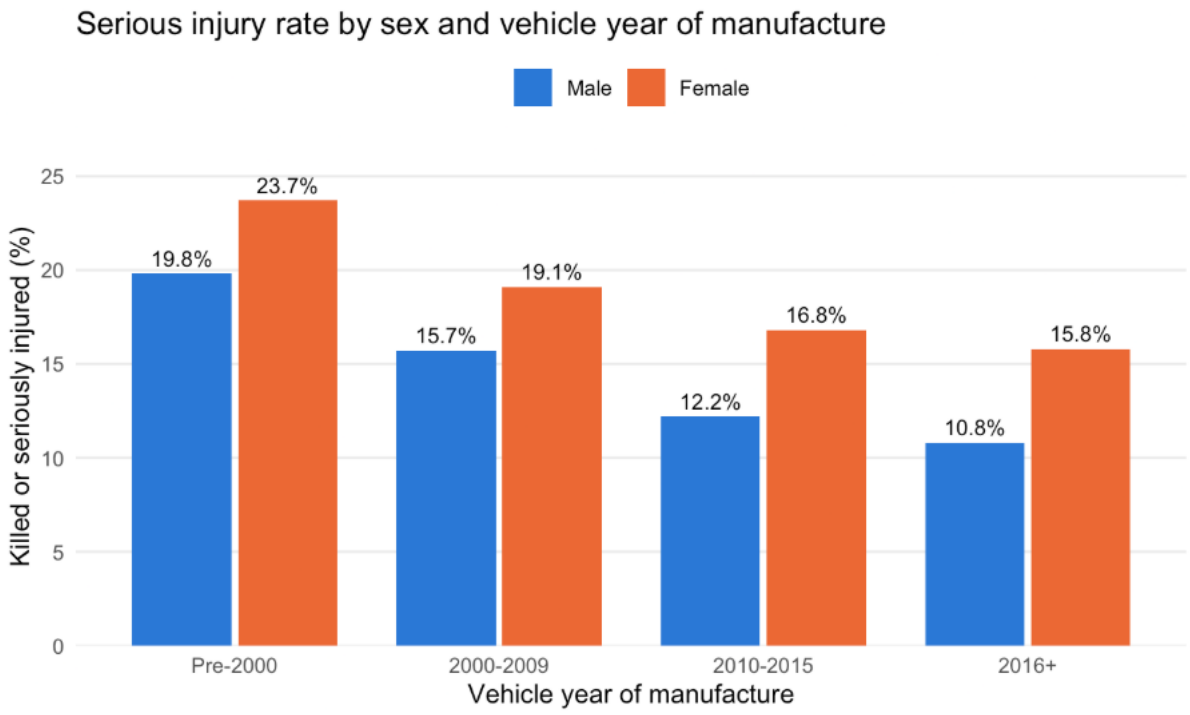


Figure 10. Observed KSI rates by sex and vehicle year of manufacture. Unadjusted rates; n = 191,464 occupants with a valid manufacture year. Source: R/03_explore.R.

Women tended to travel in lighter vehicles, with a median tare weight of 1,376 kg compared with 1,500 kg for men, and 31% of women were in the lightest quarter of vehicles compared with 19% of

men. However, women's KSI rate remained between 1.15 and 1.22 times men's within every weight group, so vehicle weight alone did not explain the difference. The observed gap was also larger for front-seat passengers (ratio 1.36) than for drivers (1.22).

The clearest pattern was the difference between types of crash (Figure 11). In multi-vehicle crashes, women's KSI rate was 16.5% compared with 11.5% for men (ratio 1.43). In single-vehicle crashes, the rates were almost identical (29.9% and 30.3%). Similarly, there was little difference in crashes into fixed objects such as trees or poles (ratio 1.01). This suggested that the gap may depend on how a crash happens, which was examined further in Research question 3.

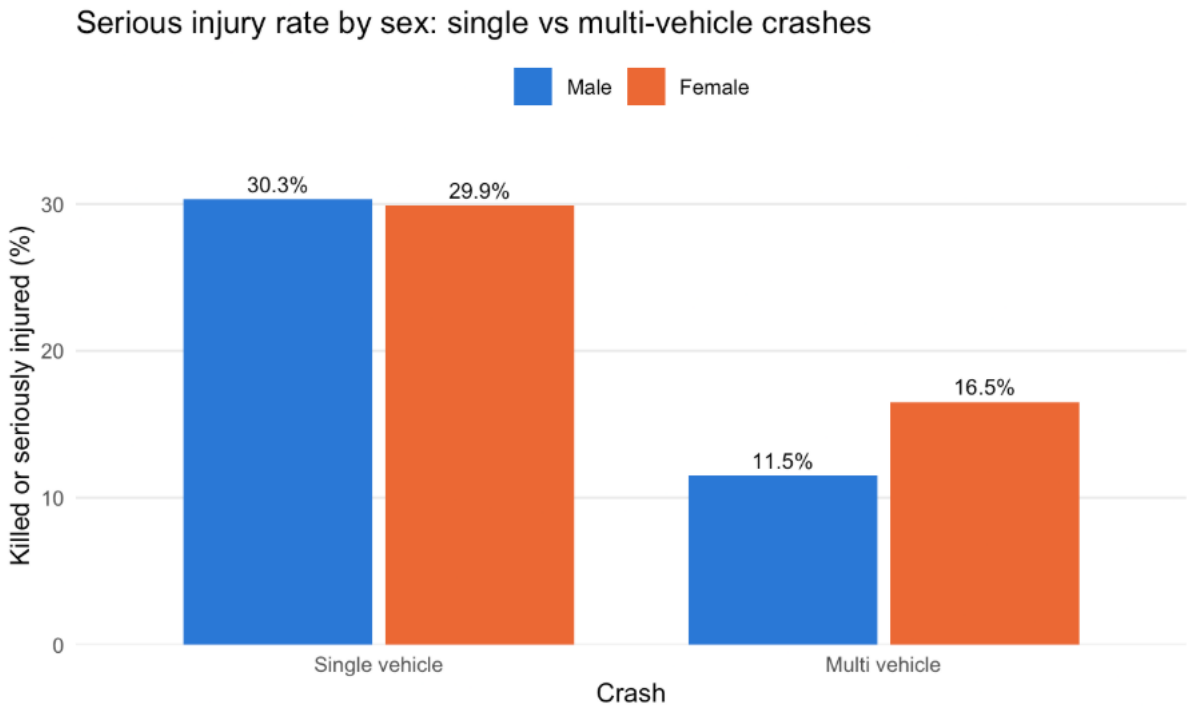


Figure 11. Observed KSI rates by sex in single-vehicle and multi-vehicle crashes. Unadjusted rates; n = 194,061. Source: R/03_explore.R.

These comparisons do not account for other factors that may differ between women and men. The full set of observed rates is provided in Table C1 (Appendix C).

Research question 1

After controlling for crash circumstances, do belted female drivers and front passengers have higher odds of being killed or seriously injured than males?

Female occupants had higher KSI odds after adjustment for the recorded crash circumstances, vehicle characteristics, seating position and occupant age. The final model estimated average KSI probabilities of 18.87% for women and 14.65% for men, giving an adjusted difference of 4.22 percentage points. The corresponding marginal odds ratio was 1.36, indicating approximately 36% higher odds in this standardised comparison.

Table 3 Overall adjusted KSI probabilities and sex difference

Measure	Estimate	95% confidence interval
Male KSI probability	14.65%	14.42% to 14.89%
Female KSI probability	18.87%	18.61% to 19.14%
Female minus male probability	4.22 percentage points	3.89 to 4.56 points

Measure	Estimate	95% confidence interval
Marginal female to male odds ratio	1.36	1.32 to 1.39

Final primary model; n = 175,453. Both sex predictions use the same reference characteristics. Intervals account for occupants sharing a crash. Source: 07_final_research_results.csv.

These findings suggest that the recorded differences in occupant, vehicle and crash characteristics did not fully account for the higher female KSI probability. However, unmeasured factors, including body size and collision severity, may still contribute to the association. Table 3 presents the overall comparison; the full set of adjusted estimates is in Appendix A.

Research question 2

How does this difference change with vehicle year of manufacture, used as a proxy for vehicle safety design era, and vehicle weight?

Newer manufacture years were associated with lower estimated KSI probabilities for both sexes. Although the primary model suggested a small increase in the absolute female-minus-male gap between manufacture years 1995 and 2020, the uncertainty interval allowed either narrowing or widening. A sensitivity model allowing a curved relationship with manufacture year estimated almost no change in the gap.

The analysis therefore did not provide robust evidence that the absolute sex gap had closed or widened with newer vehicles. Manufacture year was also only a proxy for safety design, so these findings cannot identify the effects of particular vehicle safety features.

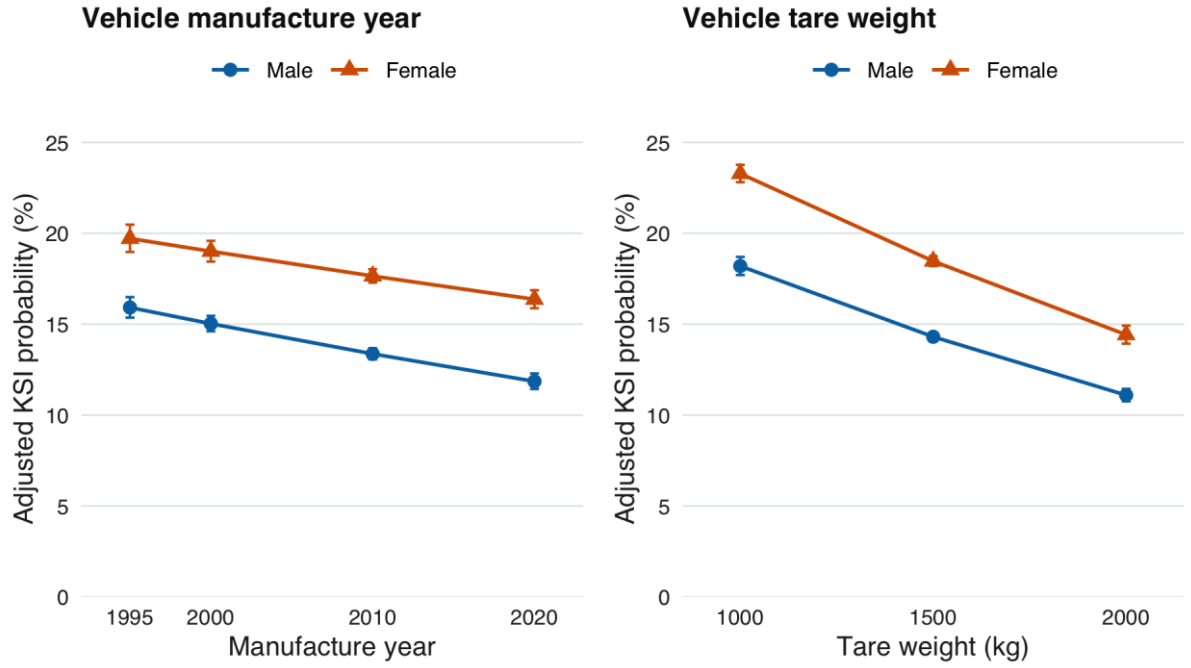


Figure 12. Adjusted KSI probabilities by manufacture year and vehicle weight

Points show final-model estimates; error bars are pointwise 95% confidence intervals. Year comparisons use the same 62,244 occupants in 2020–2024 crashes; weight comparisons use the full modelling cohort. Lines connect evaluated values. Source: 07_final_research_results.csv.

Higher vehicle weight was associated with lower estimated KSI probabilities for both sexes. The absolute sex gap decreased from approximately 5.09 percentage points at 1,000 kg to 3.32 points at 2,000 kg. However, there was little evidence that the relative sex association on the conditional odds scale varied with weight. These findings are compatible because a similar relative association can produce a smaller absolute difference when overall injury probability is lower.

The female disadvantage persisted after adjustment for vehicle weight. Figure 12 shows the probability patterns. The year-gap change was +0.71 points (95% CI −0.42 to 1.84) in the primary model and +0.02 points (−1.42 to 1.47) in the curved model. Detailed comparisons and tests are in Appendix A.

Research question 3

Does the gap hold within different crash scenarios, or is it explained by women and men having different types of crashes?

Higher female KSI probabilities remained within several crash configurations after adjustment, including rear-end, head-on, adjacent-direction intersection and opposing-direction turning crashes. Female estimates were also higher for front, side and rear impact locations. This indicates that differences in the measured crash circumstances did not fully explain the overall association.

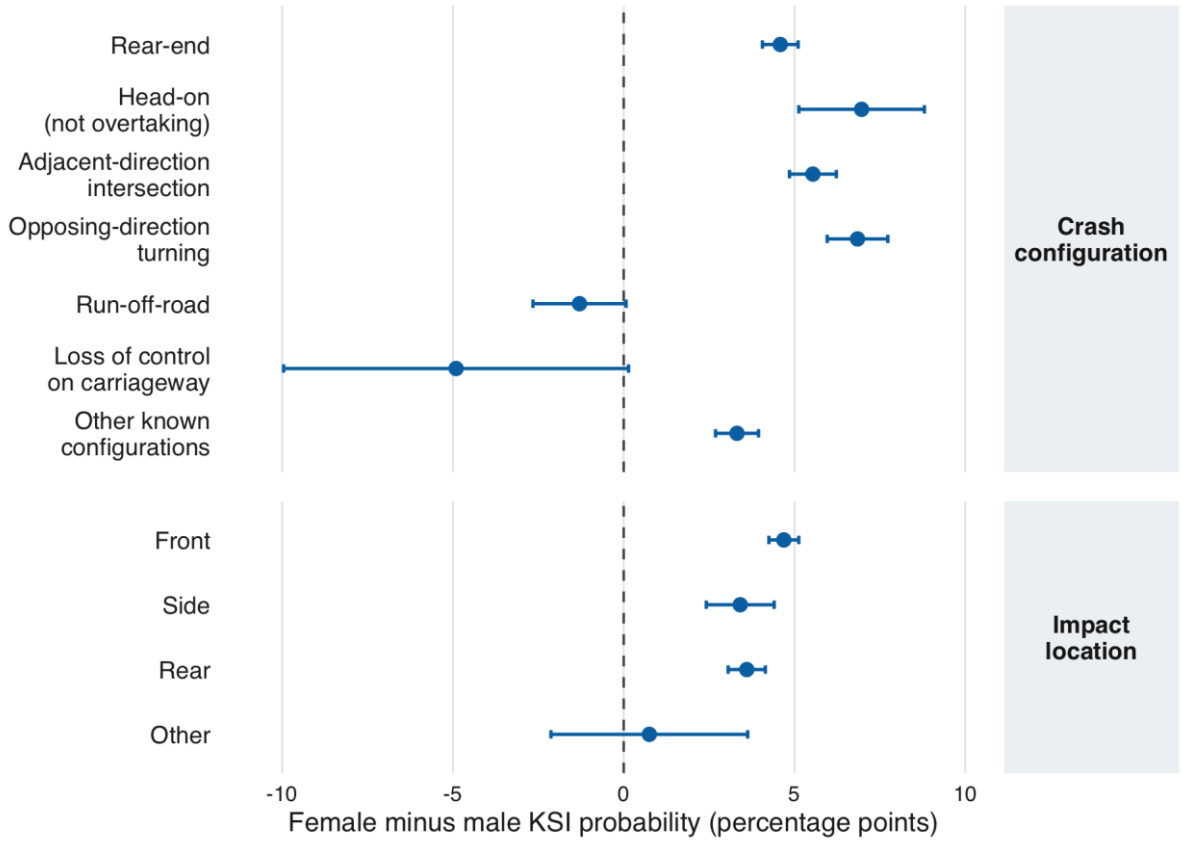


Figure 13. Adjusted female minus male KSI probability gaps by crash circumstances

Points and horizontal lines show estimates and pointwise 95% confidence intervals. Positive values indicate higher female probability; zero indicates no gap. Each configuration or impact group uses its own observed reference occupants. Source: 07_final_research_results.csv.

However, the sex difference was not consistent across all scenarios. The run-off-road and loss-of-control estimates had uncertainty intervals that included zero. These findings did not establish either a higher male KSI probability or the absence of a sex difference.

Joint interaction tests supported variation in the sex association across crash configurations and impact locations. Nevertheless, this does not mean that every pair of categories differed. Each scenario comparison also used its own observed reference population, so differences between scenarios should not be interpreted as the causal effect of changing crash type.

Figure 13 shows which intervals include zero. Numerical estimates and joint interaction tests are provided in Appendix A.

Research question 4

Does the gap differ between drivers and front-seat passengers?

The adjusted sex gap was larger among front-seat passengers than drivers. The estimated female-minus-male difference was 3.81 percentage points among drivers and 6.40 points among front passengers. The comparison between these gaps supported a larger female disadvantage among front passengers after adjustment for the recorded characteristics.

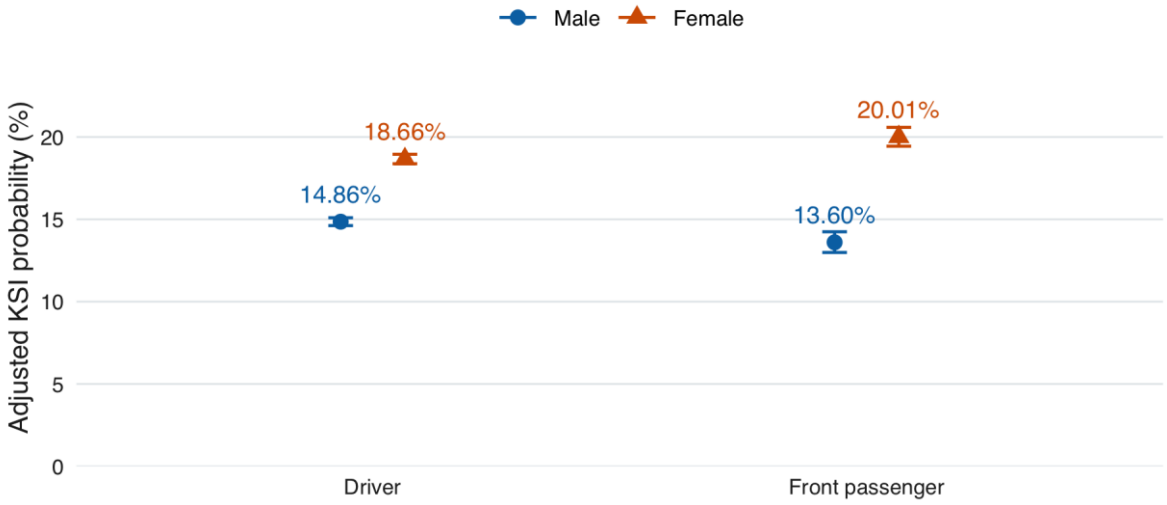


Figure 14. Adjusted KSI probabilities for drivers and front passengers

Final primary model, standardised over the full cohort. Error bars show pointwise 95% confidence intervals. The passenger-minus-driver difference in sex gaps is 2.60 points (95% CI 1.66 to 3.53). Source: 07_final_research_results.csv and 07_final_gap_changes.csv.

This finding describes a difference in the sex association across seating positions. It does not establish why that difference occurs, because information about occupant body size, restraint fit and other potentially relevant characteristics was unavailable. Figure 14 illustrates the difference in sex gaps. Full seating-position estimates and the formal comparison are in Appendix A.

Additional findings from predictive modelling

Random forest achieved slightly better predictive scores than logistic regression, although the differences were small and uncertainty intervals for those differences were not calculated. Table 4 summarises the test-set probability scores. Further performance measures, calibration charts and classification results are provided in Appendix B.

Table 4 Test performance of the predictive models

Model	ROC AUC	Average precision	Brier score	Log loss
Logistic regression	0.739	0.368	0.125	0.402
Random forest	0.743	0.380	0.124	0.399
Training prevalence	0.500	0.169	0.140	0.454

Test set: 35,305 occupants, including 5,962 KSI cases. Higher AUC and average precision are better; lower Brier score and log loss are better. Source: 07_predictive_test_metrics.csv.

Classification performance depended strongly on the probability threshold. At the default threshold of 0.5, both models achieved approximately 83% accuracy but identified only about 11% of actual KSI cases. Predicting non-KSI for every occupant would have achieved similar accuracy, demonstrating why accuracy alone was insufficient.

Thresholds selected to prioritise recall increased the proportion of KSI cases identified to approximately 80%, but only about one-quarter of positive predictions were correct. This illustrates the trade-off between identifying more serious injuries and generating more false alarms. The threshold comparisons and confusion matrices are provided in Appendix B.

These results concern retrospective prediction. The predictive analysis used earlier model specifications and does not validate the exact final association model used to answer the research questions.

Unexpected findings

Clustering mainly reproduced intersection status. The clustering analysis identified two broad groups: vehicles involved in intersection crashes and vehicles involved in non-intersection crashes. Their other most common characteristics were identical, so the clustering provided less detail about crash scenarios than anticipated. Female KSI rates were higher in both groups, but these descriptive comparisons did not establish an independent sex association. Cluster profiles and injury-rate comparisons are provided in Appendix C.

The sex gap was smaller among occupants with unknown seatbelt status. The observed gap was approximately 0.4 percentage points in this group, compared with 3.7 points among belted occupants. Including both groups reduced the overall descriptive gap to approximately 2.9 points.

The direction of the association therefore persisted, but its magnitude changed. This highlights the importance of the study's eligibility criteria and limits generalisation beyond the analysed cohort. The sensitivity results are presented in Appendix C. An adjusted analysis of the effects of these exclusions was not completed.

4. Conclusion

This study found that female occupants had higher adjusted odds of being killed or seriously injured (KSI) than male occupants in the analysed Victorian crashes. Among 175,453 belted occupants aged 16 or older, estimated KSI probabilities were 18.87% for women and 14.65% for men: a difference of 4.22 percentage points and a marginal odds ratio of 1.36. Recorded occupant, vehicle and crash characteristics therefore did not fully explain the difference, although this association does not establish causation.

The gap persisted within several crash configurations and was larger among front-seat passengers than drivers. Run-off-road and loss-of-control comparisons were inconclusive. Newer and heavier vehicles were associated with lower KSI probabilities for both sexes, but adjusting for vehicle weight did not remove the female disadvantage. Crucially, the analysis did not establish whether the gap narrowed or widened with newer manufacture years, which were only a proxy for safety design. The exploratory analysis also showed that the observed gap was concentrated in multi-vehicle crashes, with almost no difference between women and men in single-vehicle crashes.

Clustering identified broad intersection and non-intersection groups rather than detailed scenarios. Random forest offered only small predictive improvements over logistic regression, while classification involved substantial trade-offs between detecting KSI cases and generating false positives. These retrospective results do not demonstrate operational readiness.

The study lacked occupant body measurements, restraint-fit information and detailed collision-severity measures. It included only crashes in which someone was injured, and unknown seatbelt status affected the descriptive gap; adjusted exclusion sensitivity was not completed. Future research could link occupant measurements and vehicle safety ratings, assess missing-data effects and validate predictions externally. Overall, the findings indicate a persistent adjusted injury-outcome gap within this cohort, while its causes and any change attributable to newer vehicle design remain unresolved.

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Appendix A Adjusted regression results

These tables consolidate the final primary model and its matched curved-year sensitivity model. KSI means killed or seriously injured. Positive probability gaps indicate higher female estimates. All intervals are pointwise 95% intervals using crash-clustered covariance; they are not simultaneous intervals across every comparison.

Table A1 Complete primary-model standardised comparisons

Comparison	Male KSI %	Female KSI %	Gap in points (95% CI)	Marginal odds ratio (95% CI)
Overall	14.65	18.87	4.22 (3.89 to 4.56)	1.36 (1.32 to 1.39)
Year 1995	15.92	19.71	3.80 (3.08 to 4.52)	1.30 (1.24 to 1.36)
Year 2000	15.03	19.01	3.98 (3.47 to 4.49)	1.33 (1.28 to 1.37)
Year 2010	13.37	17.65	4.28 (3.93 to 4.63)	1.39 (1.35 to 1.43)
Year 2020	11.86	16.37	4.51 (3.92 to 5.10)	1.45 (1.38 to 1.53)
Weight 1000 kg	18.20	23.28	5.09 (4.42 to 5.75)	1.36 (1.31 to 1.42)
Weight 1500 kg	14.31	18.47	4.16 (3.82 to 4.50)	1.36 (1.32 to 1.39)
Weight 2000 kg	11.10	14.42	3.32 (2.75 to 3.90)	1.35 (1.28 to 1.42)
Crash Rear-end	7.48	12.07	4.59 (4.06 to 5.11)	1.70 (1.60 to 1.80)
Crash Head-on (not overtaking)	28.28	35.25	6.97 (5.13 to 8.80)	1.38 (1.27 to 1.50)
Crash Adjacent-direction intersection	12.81	18.35	5.54 (4.85 to 6.23)	1.53 (1.45 to 1.61)
Crash Opposing-direction turning	10.99	17.84	6.85 (5.96 to 7.73)	1.76 (1.63 to 1.89)
Crash Run-off-road	41.16	39.86	-1.29 (-2.66 to 0.07)	0.95 (0.90 to 1.00)
Crash Loss of control on carriageway	16.88	11.98	-4.91 (-9.96 to 0.14)	0.67 (0.44 to 1.03)
Crash Other known configurations	10.87	14.18	3.32 (2.69 to 3.95)	1.36 (1.28 to 1.44)
Impact Front	16.52	21.21	4.69 (4.25 to 5.13)	1.36 (1.32 to 1.40)
Impact Side	18.02	21.43	3.41 (2.42 to 4.41)	1.24 (1.17 to 1.32)
Impact Rear	7.66	11.26	3.60 (3.06 to 4.15)	1.53 (1.43 to 1.63)
Impact Other	17.91	18.66	0.75 (-2.13 to 3.63)	1.05 (0.87 to 1.27)
Seat Driver	14.86	18.66	3.81 (3.43 to 4.18)	1.31 (1.28 to 1.35)
Seat Front passenger	13.60	20.01	6.40 (5.56 to 7.25)	1.59 (1.49 to 1.69)

Overall, weight and seat comparisons use the full cohort of 175,453 occupants; year comparisons use the same 62,244 occupants in 2020–2024 crashes. Configuration and impact rows use their own observed groups, with identical characteristics for both sex predictions within each row. Source: 07_final_research_results.csv.

Curved year sensitivity results

Table A2 Complete curved-model standardised comparisons

Comparison	Male KSI %	Female KSI %	Gap in points (95% CI)	Marginal odds ratio (95% CI)
Overall	14.65	18.88	4.24 (3.90 to 4.57)	1.36 (1.32 to 1.39)
Year 1995	16.08	20.19	4.11 (3.22 to 4.99)	1.32 (1.24 to 1.40)
Year 2000	15.18	19.15	3.97 (3.34 to 4.60)	1.32 (1.27 to 1.38)
Year 2010	13.11	17.36	4.26 (3.77 to 4.74)	1.39 (1.34 to 1.45)
Year 2020	12.38	16.51	4.13 (3.03 to 5.23)	1.40 (1.28 to 1.53)
Weight 1000 kg	18.17	23.28	5.11 (4.45 to 5.77)	1.37 (1.31 to 1.42)
Weight 1500 kg	14.30	18.48	4.17 (3.83 to 4.51)	1.36 (1.32 to 1.39)
Weight 2000 kg	11.10	14.43	3.33 (2.75 to 3.91)	1.35 (1.28 to 1.42)
Crash Rear-end	7.48	12.07	4.59 (4.07 to 5.12)	1.70 (1.60 to 1.81)
Crash Head-on (not overtaking)	28.27	35.26	6.99 (5.16 to 8.83)	1.38 (1.27 to 1.50)
Crash Adjacent-direction intersection	12.81	18.36	5.55 (4.87 to 6.24)	1.53 (1.45 to 1.62)
Crash Opposing-direction turning	10.99	17.85	6.86 (5.97 to 7.75)	1.76 (1.63 to 1.90)
Crash Run-off-road	41.15	39.88	-1.27 (-2.63 to 0.10)	0.95 (0.90 to 1.00)
Crash Loss of control on carriageway	16.88	12.01	-4.87 (-9.93 to 0.18)	0.67 (0.44 to 1.03)
Crash Other known configurations	10.86	14.19	3.33 (2.70 to 3.96)	1.36 (1.28 to 1.44)
Impact Front	16.51	21.22	4.70 (4.27 to 5.14)	1.36 (1.32 to 1.40)
Impact Side	18.01	21.44	3.43 (2.43 to 4.42)	1.24 (1.17 to 1.32)
Impact Rear	7.65	11.27	3.61 (3.07 to 4.16)	1.53 (1.44 to 1.64)
Impact Other	17.91	18.69	0.78 (-2.10 to 3.66)	1.05 (0.87 to 1.28)
Seat Driver	14.85	18.67	3.82 (3.45 to 4.19)	1.32 (1.28 to 1.35)
Seat Front passenger	13.60	20.00	6.40 (5.56 to 7.25)	1.59 (1.49 to 1.69)

The curved model uses a four-degree-of-freedom natural spline for manufacture year with separate curves by sex. It retains the same cohort, other controls and interactions, including sex by impact. Reference populations are as in Table A1. Source: 07_final_sensitivity_results.csv.

Interaction tests and changes in the sex gap

Table A3 Joint tests of sex interactions

Model	Interaction	df	Raw p	Holm-adjusted p
Primary	Sex × manufacture year	1	0.004	0.008
Primary	Sex × weight	1	0.837	0.837
Primary	Sex × seat	1	<0.001	<0.001
Primary	Sex × configuration	6	<0.001	<0.001
Primary	Sex × impact	3	0.001	0.003
Curved	Sex × manufacture year	4	0.026	0.053
Curved	Sex × weight	1	0.866	0.866
Curved	Sex × seat	1	<0.001	<0.001
Curved	Sex × configuration	6	<0.001	<0.001
Curved	Sex × impact	3	0.001	0.004

Holm adjustment is across the five interaction tests within each model. These tests concern conditional odds, not the absolute probability-gap changes below. Source: 07_final_interaction_tests.csv.

Table A4 Changes in the standardised female minus male probability gap

Model	Comparison	Change in points (95% CI)	p
Primary	1995 to 2020	0.71 (-0.42 to 1.84)	0.216
Primary	1000 kg to 2000 kg	-1.76 (-2.81 to -0.72)	<0.001
Primary	Driver to Front passenger	2.60 (1.66 to 3.53)	<0.001
Curved	1995 to 2020	0.02 (-1.42 to 1.47)	0.975
Curved	1000 kg to 2000 kg	-1.78 (-2.83 to -0.73)	<0.001
Curved	Driver to Front passenger	2.58 (1.65 to 3.52)	<0.001

Year: 2020 minus 1995. Weight: 2,000 minus 1,000 kg. Seating: front passenger minus driver. Contrasts use unrounded estimates, so differences may not equal subtraction of displayed rounded values. Source: 07_final_gap_changes.csv.

Table A5 Supplementary curvature tests

Test	df	Wald statistic	p
Any year curvature	6	10.83	0.094
Curvature of the sex odds gap	3	3.63	0.304

Model checks: both specifications converged, using 175,453 occupants from 100,085 crashes and 29,630 KSI cases. The primary model has 46 parameters and the curved model 52. Sources: 07_final_curvature_tests.csv and 07_final_model_diagnostics.csv.

Appendix B Predictive model results

These results describe the separate crash-grouped predictive evaluation. They use earlier fitted specifications and do not validate the exact final association model in Appendix A. All test-set results are retrospective internal validation; later threshold scenarios were evaluated after the test set had already been inspected.

Table B1 Probability scores by model and sex

Model and group	n	KSI cases	AUC	Average precision	Brier	Log loss
Logistic / Male	17,624	2,613	0.760	0.360	0.111	0.364
Logistic / Female	17,681	3,349	0.712	0.374	0.139	0.440
Random forest / Male	17,624	2,613	0.766	0.374	0.109	0.359
Random forest / Female	17,681	3,349	0.712	0.385	0.138	0.438

Overall scores are in Table 4. Source: 07_predictive_test_metrics_by_sex.csv.

Table B2 Overall calibration estimates

Model	Calibration in the large	Joint intercept	Slope
Logistic regression	-0.003	-0.031	0.980
Random forest	-0.007	-0.062	0.960

Ideal calibration has intercept 0 and slope 1; overall calibration does not establish calibration within every subgroup. Source: 07_predictive_calibration.csv.

Table B3 Cross-validation scores and candidate selection

Candidate	AUC	Average precision	Brier	Log loss	Selected
Logistic linear	0.739	0.378	0.124	0.401	Yes
Logistic curved	0.739	0.378	0.124	0.401	No
RF mtry3	0.742	0.385	0.123	0.399	Yes
RF mtry6	0.737	0.379	0.124	0.402	No

Pooled out-of-fold training predictions; lowest log loss selected a candidate within each model family. Sources: 07_predictive_cv_summary.csv; R/12_predictive_evaluation.R.

Classification thresholds and error counts

Table B4 Classification performance at all four threshold rules

Model and rule	Cutoff	Recall %	Precision %	Specificity %	Accuracy %	F1
Logistic / Fixed 0.5	0.500	10.6	51.7	98.0	83.2	0.175
Logistic / Youden	0.162	66.3	30.6	69.4	68.9	0.418
Logistic / 80% recall target	0.119	79.6	25.5	52.9	57.4	0.387
Logistic / Maximum F1	0.202	56.3	35.1	78.8	75.0	0.432
Random forest / Fixed 0.5	0.500	10.6	54.8	98.2	83.4	0.178
Random forest / Youden	0.176	65.3	31.6	71.3	70.3	0.426
Random forest / 80% recall target	0.116	79.6	25.6	52.9	57.5	0.387
Random forest / Maximum F1	0.216	57.0	35.1	78.6	75.0	0.435

Thresholds were selected using training predictions and then held fixed for testing. The 80% recall target is an illustrative project scenario, not a Victorian authority requirement. Sources: 07_predictive_confusion_metrics.csv and 07_balanced_f1_test_comparison.csv.

Table B5 Confusion-matrix counts for all threshold rules

Model and rule	True positive	False positive	True negative	False negative
Logistic / Fixed 0.5	630	589	28,754	5,332
Logistic / Youden	3,955	8,990	20,353	2,007
Logistic / 80% recall target	4,743	13,832	15,511	1,219
Logistic / Maximum F1	3,354	6,215	23,128	2,608
Random forest / Fixed 0.5	633	522	28,821	5,329
Random forest / Youden	3,894	8,430	20,913	2,068
Random forest / 80% recall target	4,747	13,806	15,537	1,215
Random forest / Maximum F1	3,401	6,281	23,062	2,561

Positive = KSI. Each row totals 35,305 test occupants, including 5,962 actual KSI cases. Counts are from the same saved predictions as Table B4.

Sex specific threshold performance and predictor importance

Table B6 Performance by sex at the recall and F1 thresholds

Model	Rule	Sex	Recall %	Precision %	Specificity %	F1
Logistic	80% recall	Male	74.7	27.0	64.9	0.397
Logistic	80% recall	Female	83.3	24.6	40.2	0.380
Random forest	80% recall	Male	76.6	26.7	63.4	0.396
Random forest	80% recall	Female	82.0	24.8	42.0	0.381
Logistic	Maximum F1	Male	52.9	36.3	83.9	0.431
Logistic	Maximum F1	Female	58.9	34.2	73.5	0.433
Random forest	Maximum F1	Male	56.6	35.5	82.1	0.436
Random forest	Maximum F1	Female	57.4	34.8	74.9	0.434

Each model applies one common threshold to both sexes for a given rule. Sources: 07_recall_target_test_by_sex.csv and 07_balanced_f1_test_by_sex.csv.

Table B7 Random forest permutation importance

Predictor	Out-of-bag permutation importance
Crash configuration	0.01449
Speed zone	0.00617
Age band	0.00563
Number of vehicles	0.00441
Impact location	0.00331
Tare weight per 500 kg	0.00255
Crash year	0.00179
Manufacture year per decade	0.00108
Sex	0.00107
Seating position	0.00018

Importance describes predictive contribution within this fitted model, not causal influence or the size of an injury-risk effect. Source: 07_predictive_rf_importance.csv.

ROC and calibration plots

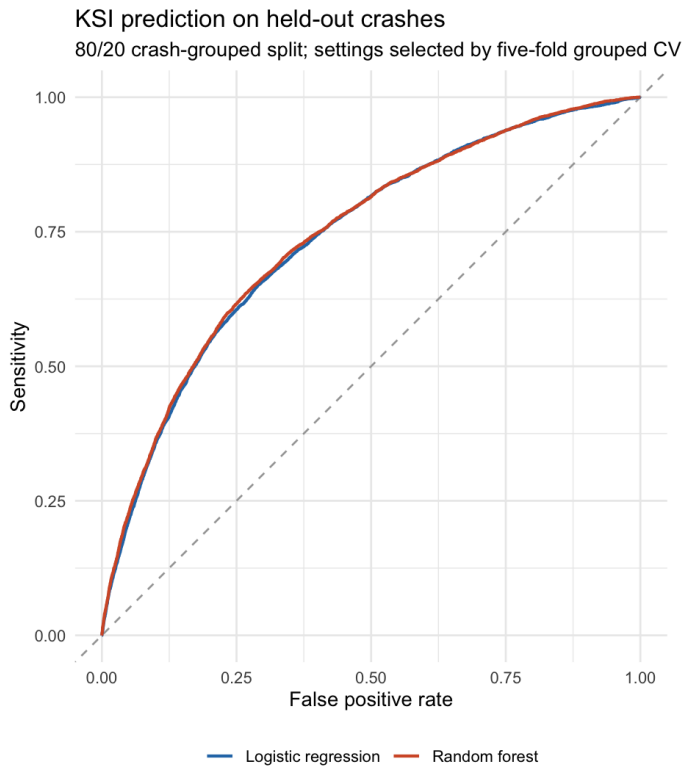


Figure B1 Test-set ROC curves for the predictive models

Higher curves indicate stronger ranking of KSI versus non-KSI outcomes. Source: R/12_predictive_evaluation.R.

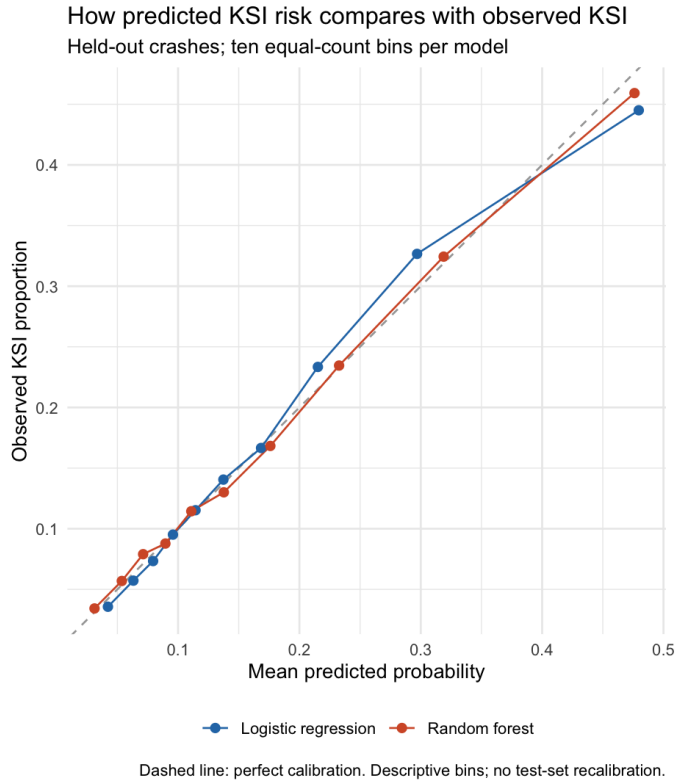


Figure B2 Test-set calibration by predicted probability group

The diagonal represents agreement between predicted and observed probability. These plots evaluate the separate predictive models, not the final association model.

Appendix C Exploratory and sensitivity results

The following results are observed, unadjusted comparisons. They use the available occupants for each factor and may therefore have different denominators. They should not be substituted for the adjusted probabilities in the main Results section.

Table C1 Observed injury rates across exploratory factors

Factor	Group	Male KSI %	Female KSI %	Gap in points	Rate ratio
Era	Pre-2000	19.8	23.7	3.9	1.20
Era	2000-2009	15.7	19.1	3.4	1.22
Era	2010-2015	12.2	16.8	4.6	1.38
Era	2016+	10.8	15.8	5.0	1.46
Weight	Q1 lightest	18.3	21.6	3.3	1.18
Weight	Q2	14.8	18.0	3.2	1.22
Weight	Q3	14.9	17.2	2.3	1.15
Weight	Q4 heaviest	12.3	14.7	2.4	1.20
Seat	Driver	14.8	18.0	3.2	1.22
Seat	Front passenger	14.6	19.9	5.3	1.36
Crash type	Collision with vehicle	11.4	16.5	5.1	1.45
Crash type	Hit fixed object	40.4	40.9	0.5	1.01
Crash type	Rollover	31.8	33.5	1.7	1.05
Crash type	Other	6.0	6.7	0.7	1.12
Impact	Front	16.4	21.0	4.6	1.28
Impact	Side	18.3	21.1	2.8	1.15
Impact	Rear	7.6	10.9	3.3	1.43
Impact	Other	17.2	19.4	2.2	1.13
Vehicles	Single vehicle	30.3	29.9	-0.4	0.99
Vehicles	Multi vehicle	11.5	16.5	5.0	1.43

Rate ratio = female rate divided by male rate. Values retain the precision of the saved exploratory outputs. Sources: the six 05_gap_by_*.csv tables.

Observed crash configuration and impact comparisons

Table C2 Observed rates by crash configuration and impact

Group	Male n	Female n	Male KSI %	Female KSI %	Gap in points	Rate ratio
Adjacent-direction intersection	20,031	21,683	12.40	18.29	5.89	1.48
Head-on (not overtaking)	5,401	4,111	27.24	35.08	7.84	1.29
Loss of control on carriageway	488	351	15.57	13.11	-2.47	0.84
Opposing-direction turning	11,677	12,523	10.50	17.97	7.48	1.71
Other known configurations	22,649	21,784	10.48	13.94	3.46	1.33
Rear-end	25,198	25,856	7.31	11.62	4.31	1.59
Run-off-road	12,646	9,600	39.49	40.54	1.05	1.03
Unknown (DCA 199)	29	34	6.90	8.82	1.93	1.28
Front	61,549	56,941	16.44	21.01	4.57	1.28
Other	1,573	1,205	17.16	19.42	2.25	1.13
Rear	20,680	23,893	7.59	10.93	3.34	1.44
Side	11,659	11,808	18.29	21.10	2.80	1.15
Unknown / not applicable	2,658	2,095	14.15	16.32	2.18	1.15

Configuration rows precede impact rows. Unknown categories are retained as reported in the exploratory outputs. These are observed rates and do not adjust for other characteristics. Sources: 06_gap_by_crash_configuration.csv and 06_gap_by_scenario_impact.csv.

Clustering selection and group profiles

Table C3 Candidate cluster solutions

Clusters	Mean category mismatches	Mean silhouette	Selected
1	1.825	Not applicable	No
2	1.373	0.364	Yes
3	1.267	0.325	No
4	1.127	0.229	No
5	1.076	0.207	No
6	0.954	0.233	No
7	0.925	0.225	No
8	0.895	0.257	No

Table C4 Profiles of the selected two clusters

Characteristic	Intersection	Non-intersection
Vehicles	83,392	69,785
Share of vehicles	54.4%	45.6%
Modal speed zone	60-70	60-70
Modal crash type	Collision with vehicle	Collision with vehicle
Modal impact	Front	Front
Road context	Intersection	Non-intersection
Modal lighting	Daylight	Daylight

Table C5 Observed KSI rates within the two clusters

Cluster	Male n	Female n	Male KSI %	Female KSI %	Gap in points
intersection	48,801	50,943	11.50	16.76	5.26
non-intersection	43,332	39,455	18.96	21.38	2.43

Selection used 10,000 vehicles; silhouettes used a 2,000-vehicle subsample. Final profiles describe 153,177 vehicles, representing 182,531 occupants. Sources: 06_cluster_k_diagnostics.csv, 06_cluster_profiles.csv and 06_gap_by_cluster.csv.

Clustering plots

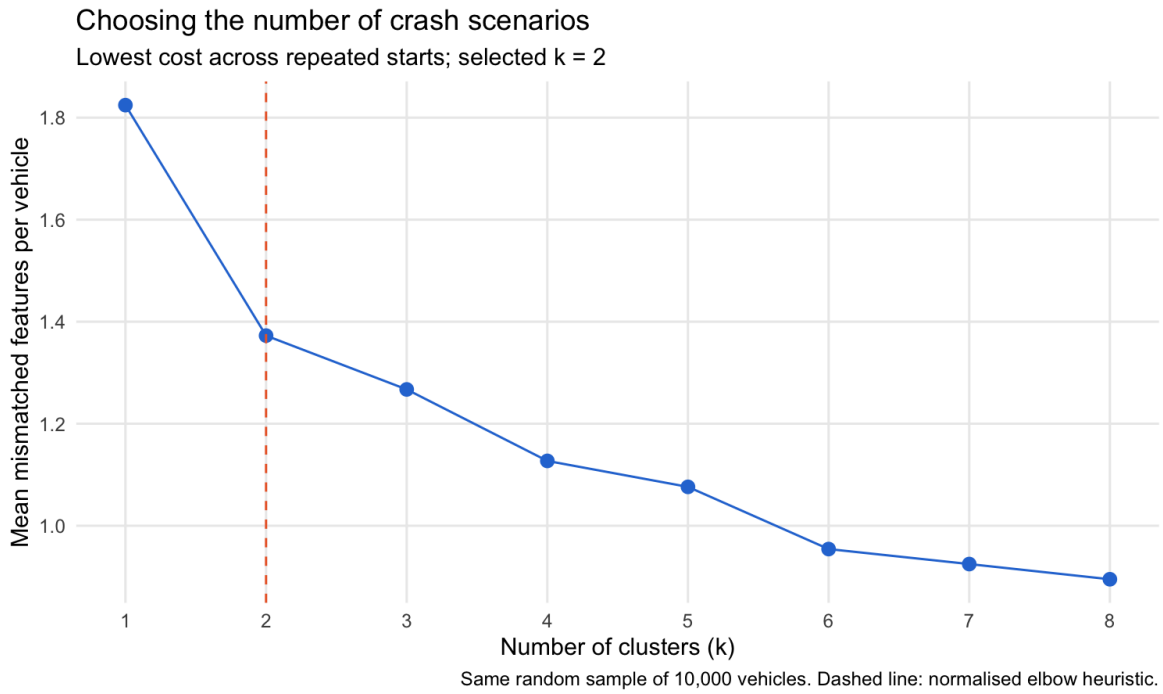


Figure C1 Selection of the number of crash clusters

The elbow heuristic selected two clusters. Source: R/04_cluster.R.

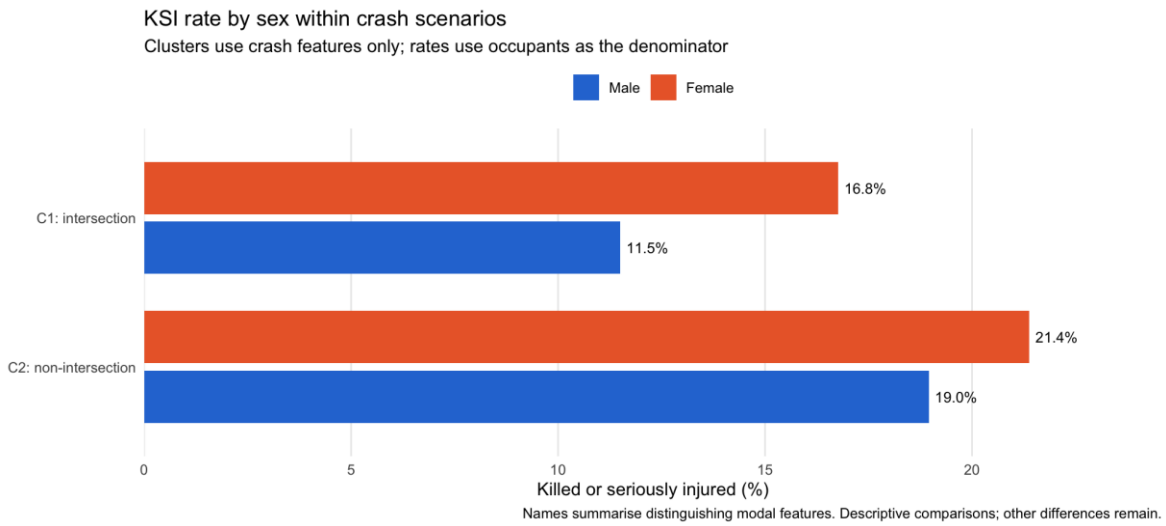


Figure C2 Observed KSI rates by sex within each cluster

Unadjusted occupant rates; clustering was fitted at vehicle level using crash circumstances. Source: R/04_cluster.R.

Seatbelt sensitivity and exploratory vehicle patterns

Table C6 Descriptive sensitivity to unknown seatbelt status

Group	Male KSI %	Female KSI %	Gap in points	Rate ratio
Seatbelt worn	14.7	18.4	3.7	1.25
Seatbelt not known	13.1	13.5	0.4	1.03
Belted and unknown combined	14.2	17.1	2.9	1.20

Combined cohort n = 273,850. Separate-group values are from 05_sensitivity_seatbelt.csv; combined rates are from the R output retained in the report's sensitivity-analysis screenshot. Values are rounded. This does not replace adjusted missing-data sensitivity.

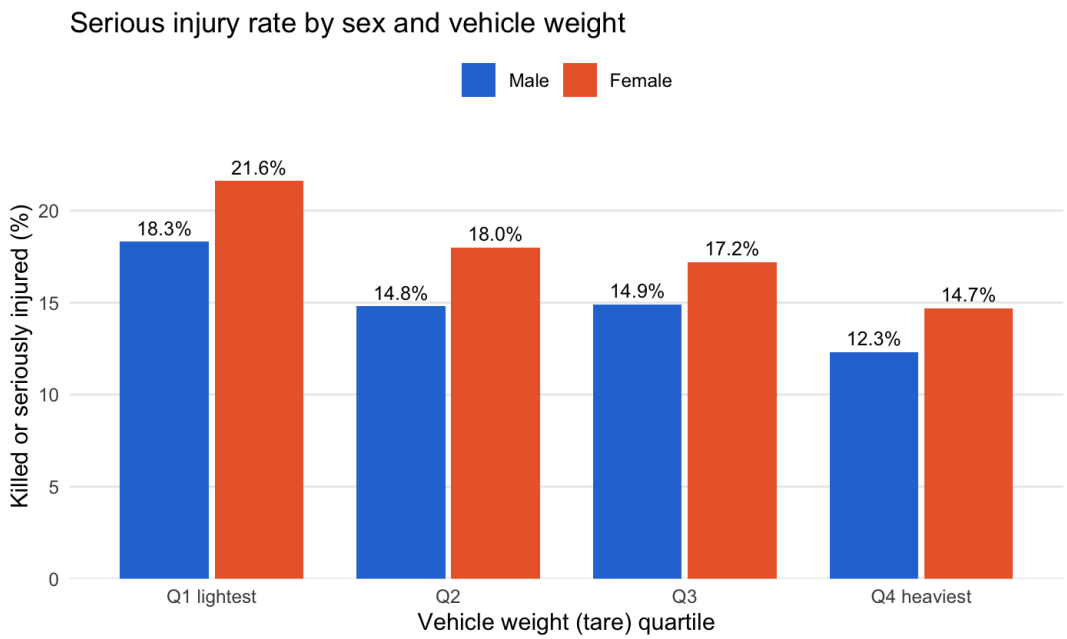


Figure C3 Observed KSI rates by vehicle weight quartile

Figure C3 shows unadjusted exploratory rates. Source: R/03_explore.R. It uses different comparisons and reference populations from the final adjusted Figure 12.

Appendix D Additional method settings

These settings supplement the concise Methodology sections. The R scripts in the project folder contain the complete implementation.

Clustering

The k-modes analysis used seed 20261102, ten starts for each multi-cluster selection fit, five starts for the full-data fit and a maximum of 30 iterations. Equal-weight category mismatches defined distance. The chosen elbow maximised distance from the line joining the endpoints of the normalised curve. Silhouette scores used a fixed subsample of 2,000 vehicles. Source: R/04_cluster.R.

Regression and standardisation

Manufacture year was centred at 2010 and scaled per decade; tare weight was centred at 1,500 kg and scaled per 500 kg. Age bands were 16–25, 26–39, 40–59 and 60+; speed bands were ≤50, 60–70, 75–90 and 100–110 km/h; vehicle count was one, two or three or more. Crash year was categorical.

Confidence intervals used HC1 covariance clustered by accident number and the delta method, treating reference characteristics as fixed. Overall, weight and seating comparisons used the full cohort. Year comparisons used the same 62,244 occupants in 2020–2024 crashes; configuration and impact comparisons used occupants within each group. The curved model retained every other control and interaction. Sources: R/06_model.R, R/16_impact_interaction.R and R/17_final_results.R.

Predictive evaluation

The crash-grouped split used seed 20261103 and produced 140,148 training and 35,305 test occupants. The random-forest grid compared three or six candidate predictors per split, with 300 trees, minimum node size 50 and minimum terminal bucket size 10. No class weighting or oversampling was applied. Spline knots were estimated within each training fit. The final selected candidates were linear-year logistic regression and the forest with three candidate predictors per split. Source: R/12_predictive_evaluation.R.

Youden's criterion maximised sensitivity plus specificity minus one. The recall rule selected the highest training cutoff achieving at least 80% recall; the F1 rule maximised the harmonic mean of precision and recall. Thresholds were frozen before their application to saved test predictions. Detailed scores and error counts are in Appendix B. Sources: R/12_predictive_evaluation.R, R/14_recall_threshold.R and R/15_balanced_threshold.R.

Appendix E Project repository

All work for this project is publicly available in our GitHub repository:
https://github.com/sylvia-griffin/Big-Data-Final-Project

The repository contains:

- **R scripts** (R/ folder): all data preparation, cleaning, exploratory analysis, clustering and modelling code, numbered in the order they should be run.
- **Data log and codebook** (docs/ folder): a record of every filtering and cleaning step with row counts, the decisions made and the reasons for them, and the meaning of each code used.
- **Figures and tables** (outputs/ folder): every chart and summary table produced by the scripts, including those not shown in this report.

- **Screenshots** (screenshots/ folder): R output captured during the analysis.
- **README:** an overview of the project, the packages required and instructions for running the scripts.

The raw crash data is not stored in the repository due to its size. It can be downloaded from the Victorian Department of Transport and Planning open data portal (https://opendata.transport.vic.gov.au/dataset/victoria-road-crash-data), or Canvas, and placed in the data/raw/ folder before running the scripts.